Visual Attention-Based Target Detection and Discrimination for High-Resolution SAR Images in Complex Scenes

Zhaocheng Wang, Lan Du, *Senior Member, IEEE*, Peng Zhang, *Member, IEEE*, Lu Li, Fei Wang, Shuwen Xu, *Member, IEEE*, and Hongtao Su

***Abstract*— The conventional methods for target detection and discrimination in high-resolution synthetic aperture radar (SAR) images usually have low accuracy and slow speed, especially for large complex scenes. To overcome these drawbacks, in this paper, we propose a target detection and discrimination method based on visual attention model. In the detection stage, to pop out the targets and suppress the background clutter in the saliency map, we select the task-dependent scales from the Gaussian pyramid of the original SAR image. Moreover, we adopt the clustering algorithm to remerge several isolated focus of attention areas, which are obtained from the saliency map, into a complete target region. The candidate target SAR image chips are extracted with relative high accuracy and low time cost in this stage. Since there may be single target, multiple targets, or partial targets with complex clutter in each SAR image chip, it is hard to acquire accurate target-shaped blob via segmentation. Some classical discrimination features which are extracted based on target segmentation may lose effectiveness. In the discrimination stage of our method, to solve the above problem, based on the saliency and gist (SG) features for optical satellite images, we propose the modified SG (MSG) features for SAR target discrimination. The MSG features are complementary to each other and can provide a more complete description of the extracted SAR image chips without segmentation, which also reduces the computation burden. The experimental results on the synthetic images and miniSAR real SAR image data set demonstrate that the proposed target detection and discrimination method can detect and discriminate the targets from the complex background clutter with high accuracy and fast speed in high-resolution SAR images.**

***Index Terms*— Complex scenes, synthetic aperture radar (SAR), target detection, target discrimination, visual attention.**

I. Introduction

WITH the overwhelming amount of high-resolution synthetic aperture radar (SAR) images available today, it is becoming increasingly desirable to develop the fast and reliable techniques for SAR automatic target recognition (ATR) in the military and civilian fields.

Manuscript received January 4, 2017; revised June 26, 2017 and October 13, 2017; accepted October 17, 2017. Date of publication November 21, 2017; date of current version March 23, 2018. This work was supported in part by the National Science Foundation of China under Grant 61771362, Grant 61322103, Grant 61231017, and Grant 61671354, and in part by the Science Foundation of Shaanxi Province under Grant 2015JZ016. *(Corresponding author: Lan Du.)*

The authors are with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China, and also with the Collaborative Innovation Center of Information Sensing and Understanding, Xidian University, Xi'an 710071, China (e-mail: dulan@mail.xidian.edu.cn).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2017.2769045

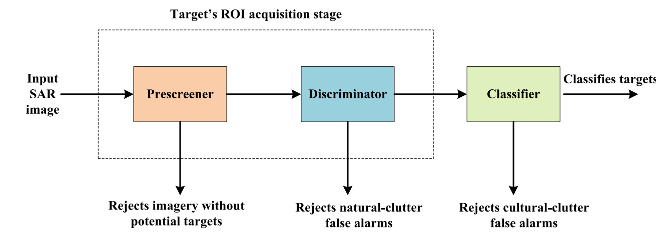


Fig. 1. Block diagram of the SAR ATR system developed by the Lincoln laboratory [1]–[4]. The detector locates candidate targets based on the brightness of pixels in the input SAR image, the discriminator rejects natural-clutter false alarms, and the classifier rejects nontarget cultural-clutter and classifies targets by target type.

The most popular SAR ATR system is developed by the Lincoln Laboratory [1]–[4] which consists of three stages: detection, discrimination, and classification (see Fig. 1). In the detection stage, a two-parameter constant false alarm rate (CFAR) [6]–[8] detector selects candidate target chips (small SAR images contain targets or clutter) in an SAR image by examining the amplitude of the radar signal in each pixel of the SAR image, in which the true targets as well as numerous clutter false alarms, such as trees, buildings, streetlights, and street edges, are extracted as the candidate target chips. Target discrimination, essentially as a binary classification problem, is a postprocessing for the candidate target chips extracted from the detection stage, with the purpose of reducing classification/recognition cost, by removing natural clutter. After the detection and discrimination stages, target's region of interest (ROI) chips are obtained, then these ROI chips are input into the classification stage, where a 2-D pattern matching algorithm rejects the cultural clutter (i.e., man-made objects that are not targets of interest) and classifies the remaining detections by target type (tank, armored personnel carrier, or howitzer). There are numerous existing studies focusing on the classification stage [31]–[33], [36]–[39] and they have obtained good classification performance. However, many problems have still not been solved before the classification stage, i.e., the detection and discrimination stages. Moreover, due to the foundation and the importance of the detection and discrimination stages (they are also called the target's ROI acquisition stage) for the SAR ATR system, this paper focuses on these two stages.

A. Detection Stage

As described above, CFAR is a conventional target detection method which has been applied into several real SAR ATR systems [3], [9], e.g., SAIP system and An/APG-76 system. However, it has some disadvantages in practice. First, the proper clutter statistical model is difficult to choose when the scene is complex, such as urban areas, and an inappropriate model will result in a large number of false alarms or missed alarms. Second, the speed is too slow to be applied into the real-time SAR ATR system, due to the sliding window scheme in the whole SAR image.

Considering the above-mentioned issues, an excellent target detection method for SAR images should have the following properties.

1) *High Detection Accuracy:* The detection algorithm should have high detection rate and low false alarm rate not only for simple scenes but also for complex scenes. The high detection rate guarantees that most targets are detected, while the low false alarm rate alleviates the workload of the following procedures.
2) *Independent on the Clutter Statistical Model:* Since the statistical property of the clutter in a complex scene varies, it is difficult to determine a fixed clutter statistical model in the detection algorithm. A good detection algorithm should be robust and independent on the clutter statistical model; otherwise, it will cause low detection accuracy.
3) *High Detection Speed:* Facing the large scene in the real SAR image, the detection algorithm should be implemented as quickly as possible; otherwise, it will affect the efficiency of the whole SAR ATR system, especially for military application.

Recently, it is noted that the ability of human visual attention system to detect targets from the complex scenes in optical images is extraordinarily fast and reliable [10]–[12], and numerous excellent computational visual attention models have been proposed to imitate the architecture of human vision system.

Itti *et al.* [10] proposed a biologically inspired visual attention model where the image saliency is measured by center-surround contrasts. This model is independent on the statistical characteristic of the image and has simple calculation. However, Itti's model is purely data driven and does not consider the detection task. From the point of view of information theory, Bruce and Tsotsos [13] presented the attention based on information maximization model which uses Shannon's self-information (SI) measure for calculating the saliency of image regions. This model needs a large number of image patch samples to measure the SI of image regions. However, it is not easy to acquire a large number of samples for SAR image in the target detection stage. Then, using Bayesian theory, Itti and Baldi [14] defined the saliency as "Bayesian Surprise" which is realized by computing the Kullback–Leibler (KL) divergence between posterior and prior beliefs of an observer. It requires assuming specific statistical model of the image for calculating the Bayesian Surprise. As mentioned above, we hope that the target detection method for SAR image is independent on the statistical model. Therefore, this model is also not fit for dealing with our target detection problem.

All the above-mentioned saliency definitions are based on local contrast. Meanwhile, some other algorithms focus on constructing the saliency based on global contrast. From the perspective of spectral domain, a spectral residual (SR) approach based on the Fourier transform was proposed by Hou and Zhang [15], where they analyzed the log spectrum of an input image and extracted the SR of an image in spectral domain to construct the saliency map. Then, Yu *et al.* [16] investigated the relationship between the sign of the discrete cosine transform coefficient and the visual saliency, and proposed the pulsed cosine transform (PCT) visual attention model. Though these two models are rather simple, they usually have low accuracy in application, since they do not use any prior information of the target of interest. Recently, Cheng *et al.* [17] proposed a global regional contrast-based saliency detection algorithm. This algorithm produces high-quality saliency maps, which can uniformly highlight the entire object regions rather than just pop out the edge of the object as the local contrast method does. However, when calculating the saliency value of a region, it needs to compare the region with all the other regions in the whole image, which is rather time consuming.

As mentioned above, numerous computational visual attention models are now available in the field of the optical images, which have demonstrated successful applications in target detection, mobile robotics, and smart advertising.

In recent years, the idea of visual attention has been introduced into SAR target detection and obtains some good performance [18]. Yu *et al.* [18] used the above-mentioned PCT visual attention model to detect the ship in SAR images. To guarantee the detection rate of weak targets in SAR images, Liu *et al.* [19] modified Itti's model and proposed a singular value decomposition (SVD)-based visual attention detection algorithm for SAR image. However, the computation speed of SVD is very slow especially for the large scene of the SAR images. In addition, the above two methods are purely data driven; therefore, they cannot yield good performance in complex scenes. According to the prior knowledge that ships are in the water region rather than on the land, Hou *et al.* [20] proposed a ship detection method in SAR images based on visual attention model to improve the ship detection performance.

The above-mentioned visual attention-based target detection methods for SAR images are effective in some simple scenes, such as ship detection in water regions. If the task is to detect the targets in complex scenes, such as detecting vehicle targets in urban areas with clutter such as trees and buildings, they may lose effectiveness. Among these methods, Hou *et al.* [20] introduced the prior knowledge into the ship detection in water regions and improved the detection performance. Therefore, it is inspiring to introduce suitable prior information into the visual attention model to improve the target detection accuracy in complex scenes.

To detect the targets from the complex scenes and overcome the defects of CFAR, based on the classical visual attention model, Itti's model, we propose a novel target detection method for SAR images. According to the prior knowledge of targets to be detected, task-dependent scales are selected from the Gaussian pyramid of the original SAR image, to pop out the targets and suppress the background clutter in the saliency map. The salient pixels are obtained with the operation called "shifts of the focus of attention (FOA)" in the saliency map, and the clustering algorithm is followed to remerge the salient pixels. Finally, some false alarms are eliminated according to the number of pixels of every cluster.

B. Discrimination Stage

The classical target discrimination features for single-channel SAR images include the old Lincoln features [4], [24], new Lincoln features [25], Gao's features [26], [27], and Bhanu's features [21], [28]. The old Lincoln features are composed of the texture features, shape features, contrast features, and contiguousness features. The new Lincoln features are made up of the spatial boundary features. The Gao's features contain the signal-to-noise-ratio features. The Bhanu's features consist of the distance features, projection features, and moment features. The old Lincoln features and the Bhanu's features are extracted mainly based on the target segmentation results, while the new Lincoln features and the Gao's features are extracted based on the target boundary. Some of these classical discrimination features have been utilized into the real SAR ATR system [3], [9], and the performance of them is impressive in some simple scenes, such as the moving and stationary target acquisition and recognition public release data set [21], [26]–[28]. Nevertheless, they have several disadvantages in practice. First, all the classical discrimination features are specially designed for the target chips containing only one complete target and the clutter chips being relatively simple. However, for some complex scenes, such as the miniSAR real SAR image data set that is used in this paper, there are single target, multiple targets, and partial target with complex clutter cases in the target chips obtained by the detection stage. Second, some classical discrimination features which are extracted based on target's shape and size, need segmentation preprocessing to obtain the target-shaped blob. However, the segmented target-shaped blob usually varies with the segmentation methods, and thereby, the corresponding features are not robust especially for complex scenes. Third, the extraction procedures of some classical discrimination features are complex, such as requiring segmentation preprocessing, and the extraction speed is too slow for the real-time ATR system.

Considering the above-mentioned issues, an excellent target discrimination feature set for SAR images should have the following properties.

1) *High Discrimination Accuracy:* We expect the high discrimination accuracy not only for some simple scenes but also for some complex scenes.
2) *Independent on Segmentation Preprocessing:* As discussed above, some classical discrimination features based on target-shaped blob are highly dependent on the segmentation results and not robust for complex scenes. To obtain more robust features, the new discrimination features should be extracted without segmentation preprocessing.
3) *High Feature Extraction Speed:* The discrimination features should be extracted quickly without complex extraction procedures.

Li and Itti [11] proposed two complementary sets of visual attention-based features, saliency and gist (SG) features, which are initially designed to detect the targets in optical satellite images. The saliency feature channel includes ten feature subchannels: intensity, orientation (0°, 45°, 90°, and 135°, combined into one "orientation" subchannel), local variance, entropy, spatial correlation, T-junctions, L-junctions, X-junctions, endpoints, and surprise. The gist feature channel includes 18 subchannels: intensity, four orientations (0°, 45°, 90°, and 135°), and four L-junctions (0°, 45°, 90°, and 135°), four T-junctions (0°, 45°, 90°, and 135°), four endpoints (0°, 45°, 90°, and 135°), and X-junction. For convenience, we refer to the subchannels related to the junctions, i.e., T-junctions, L-junctions, X-junctions, and endpoints for the SG features, as junction subchannels. Here, the saliency feature channel exploits the intensity and spatial distribution of those objects within a scene which stand out by being significantly different from their local neighbors, while the gist feature channel focuses on summarizing the global statistics and contextual information over the entire scene, and they are combined together to give a complete description of the scene.

In this paper, we introduce the SG features to solve the target discrimination problem in SAR images. The SG features are complementary to each other and can provide a more complete description of the image chip without segmentation, which also reduces the computational complexity.

Due to the speckle noise, the pixels of SAR images are greatly fluctuant even in homogenous regions. Therefore, the SG features cannot be directly used into SAR target discrimination. In this paper, based on the SG features in [11], we propose the modified SG (MSG) features fitting for SAR target discrimination.

In this paper, we find that the subchannels related to the junctions, i.e., T-junctions, L-junctions, X-junctions, and endpoints, are almost unuseful for SAR target discrimination. Moreover, for high-resolution SAR images with sophisticated statistical characteristics, it is not easy to assume the appropriate prior distribution for the surprise subchannel. Therefore, we exclude the junction and surprise (JS) subchannels from the SG features in the MSG features. In addition, we add three new subchannels which are fit for SAR target discrimination into the MSG features. Specifically, we add the SI, SR, and gratitude magnitude (GM) into the saliency channel.

The remainder of this paper is structured as follows. Section II introduces the novel target detection method based on visual attention model. Section III describes the new target discrimination method using the MSG features. Section IV shows the details of our experiments and gives the result analysis. Finally, the conclusion is given in Section V.

II. Target Detection

The flowchart of the proposed target detection method for SAR images is given in Fig. 2. An example of a small

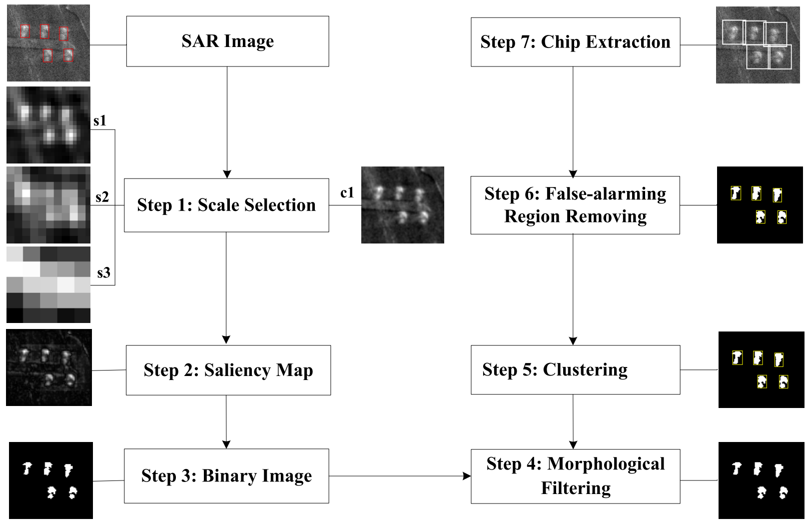


Fig. 2. Flowchart of the proposed target detection method for SAR images.

SAR image is shown along each step for the better illustration purpose. The scene of the SAR image used here is rather simple, and more complicated scene will be employed in the experimental section. In the first image, the ground truths of the five targets in this example are marked by red rectangles. A brief explanation for each step is given in the following.

Step 1: In order to pop out the targets of interest and obscure the background clutter, we select the task-dependent scales from the Gaussian pyramid in step 1 of our method, rather than using the fixed scales for the operation of center surround as Itti did. Here, we show our selected images from the Gaussian pyramid of our example.

Step 2: Based on the result of the scale selection in step 1, we combine all the feature maps obtained by the operation of center surround to construct a saliency map.

Step 3: Since the targets are brighter than most clutter in the saliency map via scale selection, the targets should have priority to be detected. In step 3, a binary image marking all the salient pixels is obtained via the operation called "shifts of the FOA" [10] in the saliency map.

Step 4: In order to connect the narrow gaps and delete some small false alarms, we apply the morphological filters to the binary image.

Step 5: Due to the speckle noise in SAR images, the salient pixels belonging to a target are usually separated into several parts in the binary image. We adopt the clustering method proposed by Gao *et al.* [7] to connect several parts into a complete target region, rather than regarding every salient region as one complete target as Itti *et al.* [10] do. The salient pixels are clustered into five categories in this example and they are marked by yellow rectangles.

Step 6: After the clustering processing, some large regions still exist in the binary image, which are not the targets of interest obviously. In fact, the number of target pixels in the SAR image is usually smaller than that of the real target of interest; thus, the number of target pixels in the binary image after clustering has a superior value $S_{\max}$. In order to eliminate the false alarms, the clustered regions with the number of pixels S larger than that corresponding to the maximum number of pixels $S_{\max}$, i.e., $S > S_{\max}$, are not treated as the target cluster in the next step. In this example, the number of pixels in each cluster in step 5 is just within the threshold range; therefore, all the clusters are regarded as potential targets.

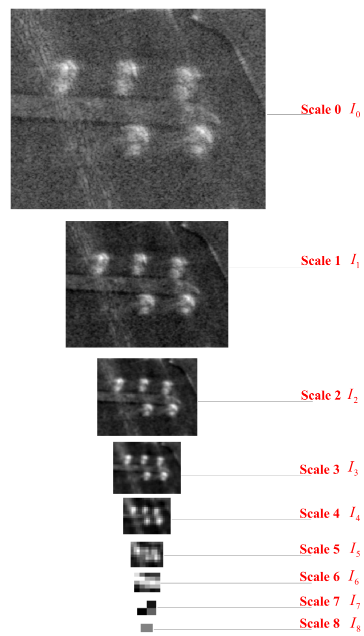


Fig. 3. Gaussian pyramid of an SAR image example.

Step 7: The candidate target chips are extracted from the original SAR image based on the centroids of target cluster in the binary image, and in this example, they are marked by white rectangles.

Among the above steps, steps 1, 3, 5, and 6 are important for the proposed target detection method, and we will give their detailed descriptions in the following.

A. Scale Selection

Fig. 3 shows the Gaussian pyramid of an SAR image example. The nine images in different spatial scales are created by progressively low-pass filtering and subsampling the input SAR image with the reduction factors 1:1 (scale 0, original scale), 1:2 (scale 1), 1:4 (scale 2), ..., 1 : 256 (scale 8) for both of horizontal and vertical dimensions. In this example, we get nine multiscale SAR images and they are expressed as $I_\sigma = \{I_0, I_1, I_2, \ldots, I_8\}$. Itti *et al.* [10] used an operation called center surround which is implemented as the difference between the center pixel at fixed scales $c \in \{2, 3, 4\}$ in the Gaussian pyramid and the corresponding surround pixel at fixed scales $s = c + \delta$ with $\delta \in \{3, 4\}$, i.e., $I_2 - I_5$, $I_2 - I_6$, $I_3 - I_6$, $I_3 - I_7$, $I_4 - I_7$, and $I_4 - I_8$, for different images with different detection tasks. However, we find that at some scales, the large area clutter, e.g., trees, grasslands, and buildings, is still clear, while the small vehicle targets are fused into the background in the SAR image. Therefore, the appropriate scale selection becomes a crucial step for target detection in SAR images. In our proposed detection method, the surround and center scales are employed to exclude some large and small clutter which is obviously not the targets of interest. In the following, we explain how to select the surround and center scales.

First, the surround scales s are used to exclude some large clutter, and they should ensure the targets of interest coarse enough while the background clutter is still unambiguous, so that the targets of interest can be highlighted by the differences between the image of center scale and the images of surround scales, i.e., the center-surround [10] operation. To achieve this goal, the surround scales s are selected based on the detection task and the resolution of the SAR image. Assume that the prior maximum length and width of the targets of interest are L and W, respectively. Both the range and azimuth resolution of the SAR image are ΔR. Note that in this paper we are interested in the target detection problem in high-resolution SAR images, and ΔR should be small enough so that the targets of interest in the image are the extended targets, i.e., one target contains multiple pixels. Moreover, we assume that the slant range SAR image has been transformed to the ground range SAR image [43] via the geometric correction in the postprocessing of SAR imaging. Then, the maximum length l and the width w of the targets in the SAR image are

$$l = L/\Delta R \tag{1}$$

$$w = W/\Delta R. \tag{2}$$

If we define

$$M = \min(l, w) \tag{3}$$

and then the best surround scale s_{best} should be

$$s_{\text{best}} = \text{round}\left(\log_2^M\right) \tag{4}$$

where round(·) represents the operation of rounding. Accordingly, the objects whose lengths or widths are smaller than $2^{s_{\text{best}}}$ can be highlighted via the operation of center surround. Since the size prior information of the targets may be not accurate enough, several scales close to s_{best} can also be selected as the surround scales, which can make the proposed detection algorithm more robust.

Second, the center scale c is used to exclude some small clutter, and it should be selected from $0 \sim (\min\{s\} - 1)$.

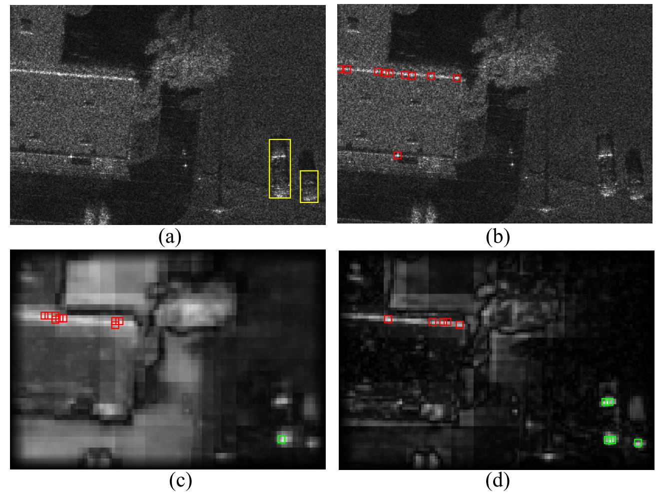


Fig. 4. Comparison of the first ten attended locations in the order of decreasing saliency using different methods. (a) Original SAR image (the vehicle targets are marked by yellow rectangles). (b) Result via using the original image as the saliency map. (c) Result via the saliency map with the scales in [10] ($c \in \{2, 3, 4\}$ and $s = c + \delta$ with $\delta = \{3, 4\}$). (d) Result via the saliency map with our selected scales ($c \in \{2\}$ and $s = \{4, 5, 6\}$). In (b)–(d), the attended locations of the vehicle targets are marked by the green rectangles and those of the clutter are marked by the red rectangles.

The larger the c, the larger can be excluded the clutter, and vice versa. Since the target occlusion problem may exist in some azimuth angle in the SAR image, where the target size may be a little smaller than the real target size, in our proposed detection method, we select the center scale to be smaller than the scale corresponding to the real target size. If the center scale is too large, some targets with occlusion will be excluded as the clutter, while the small center scale can avoid excluding this small clutter and retain the targets with occlusion.

Based on the selected center and surround scales, several feature maps can be obtained by the center-surround operation, and then a saliency map I_S is obtained by linearly fusing these feature maps.

Fig. 4 is employed as an example to show the advantage of our scale selection method using the first ten attended locations in the order of decreasing saliency. Fig. 4(a) is a piece of real SAR image with complex scene, in which the two vehicle targets are marked by yellow rectangles. In Fig. 4(b)–(d), the attended locations of vehicle targets are marked by the green rectangles and those of the clutter are marked by the red rectangles. As shown in Fig. 4, the vehicle targets are more salient in the saliency map using our scale selection method than those in the saliency maps using the original image and the fixed scales method.

B. Generation of Binary Image

The maximum value of the saliency map A_m indicates the most salient location, to which FOA should be directed. At an FOA location, seeded region-growing segmentation algorithm [22] is employed to generate a salient region around the FOA location. To make sure that the salient region would not be focused again, the current salient region is set to be zero in the saliency map I_S. Then, the maximum value of the saliency map I_S will become smaller and smaller with shifts of the FOA. Here, the termination condition for shifts of the FOA is

$$A_m < T, \quad A_m = \max(I_S). \tag{5}$$

In (5), the threshold T is calculated as follows. In the original SAR image, some strong clutter, such as the building clutter, may have larger intensities than that of the targets of interest. Nevertheless, since the building clutter usually has the larger size than that of the targets of interest, i.e., the vehicle targets, in this paper, the intensities of the building clutter pixels can be suppressed by the proposed detection method in the saliency map via the task-dependent scale selection. Therefore, the intensities of the target pixels should be larger than those of the clutter pixels in the saliency map of our method, and the tailed part of the histogram of the saliency map represents the target pixels. Referring to [7], T can be determined adaptively according to the histogram of the whole saliency map. Let F be the cumulative distribution function obtained from the histogram of the saliency map I_S. Then, T can be obtained from

$$1 - F(T) = \alpha \tag{6}$$

where $\alpha \in [0, 1]$ is an empirical value that indicates the confidence of a pixel being a target pixel, since the targets of interest are highlighted in the saliency map. Moreover, due to the targets of interest, such as the vehicles in Fig. 2, only contain a small proportion of pixels in the whole SAR image, α should be small. Here, $\alpha = 0.515\%$ in Fig. 2.

"Shifts of the FOA" is implemented in the order of decreasing saliency in the saliency map until the maximum value of the saliency map A_m satisfies (5). After that, we obtain a binary image I_B which marks all the salient regions.

C. Clustering

The clustering algorithm proposed by Gao *et al.* [7] clusters every two connected regions with their maximum pixel distance within a threshold $d_{\max}$. According to (1) and (2), the threshold $d_{\max}$ is calculated as

$$d_{\max} = \sqrt{l^2 + w^2}. \tag{7}$$

D. False Alarm Elimination

In practice, to eliminate false alarms, the maximum number of pixels of the targets $S_{\max}$ is represented as

$$S_{\max} = L \times W/(\Delta R \times \Delta R). \tag{8}$$

E. Comparison With Itti's Model

There are mainly two differences between the proposed detection method and the Itti's model. First, to highlight the targets and suppress the background clutter in the SAR image, we select the task-dependent scales from the Gaussian pyramid based on the size prior information of the targets, rather than using the fixed scales for the operation of center surround as Itti *et al.* [10] did. Second, to join the several isolated FOA areas into a complete target region, we adopt the clustering method proposed by Gao *et al.* [7], instead of regarding every isolated FOA area as one complete target in the Itti's model.

III. Target Discrimination

In this section, based on the SG features for optical images, we propose the MSG features which are fitting for SAR target discrimination.

The MSG features are combined together to form the MSG feature vector. Then, the MSG feature vector is sent to the linear support vector machine (SVM) to realize the discrimination task. In this paper, we use the library for SVM [23] implementation for the linear SVM.

A. Comparison With the SG Features

Fig. 5 shows the block diagram of feature extraction procedures of the proposed MSG features for SAR images and the SG features for optical images. Here, the red solid lines and the blue solid lines correspond to the MSG features for SAR images, and the red solid lines and the black dotted lines correspond to the SG features for optical images, i.e., the MSG features exclude the features denoted by the black dotted lines from the SG features according to the characteristics of SAR images and add the new features denoted by the blue solid lines. The reasons for removing the features which are denoted by the black dotted lines in Fig. 5 are as follows.

1) Reasons for Excluding the Junction Subchannels: The junction subchannels include four subchannels: L-junction, T-junction, X-junction, and endpoint, and they describe the local edge structures of image chips. For target detection in optical images [11] with smooth intensities, the targets of interest should have the local structures of T-junctions, L-junctions, X-junctions, and endpoints, while the background clutter do not have these local structures. However, for SAR images, due to the strong random fluctuation caused by the speckle noise, both the targets of interest and the background clutter may have some junction responses. Here, the junction responses of both the targets of interest and background clutter may be generated by the random speckle noise rather than the object-specific structures, and the junction features may lose their effectiveness in target discrimination for SAR images. Therefore, we exclude the junction subchannels from the original SG features in our proposed MSG features.

2) Reasons for Excluding the Surprise Subchannel: Briefly, surprise quantifies the difference between prior and posterior beliefs of an observer as new data are observed [14]. If observing new data causes the observer to significantly reevaluate his/her/its beliefs about the world, then observation will cause high surprise. Specifically, surprise is calculated as

$$\begin{aligned} \mathrm{Surp}(D, M) &= \mathrm{KL}(P(M|D), P(M)) \\ &= \int_M P(M|D) \log \frac{P(M|D)}{P(M)} \end{aligned} \tag{9}$$

where D is the data observation, M is the belief model of the observer, $P(M)$ is the prior distribution of the belief model, $P(M|D)$ is the posterior distribution of the belief model, and $\mathrm{KL}(\cdot)$ is the KL divergence [14] of two distribution. For optical

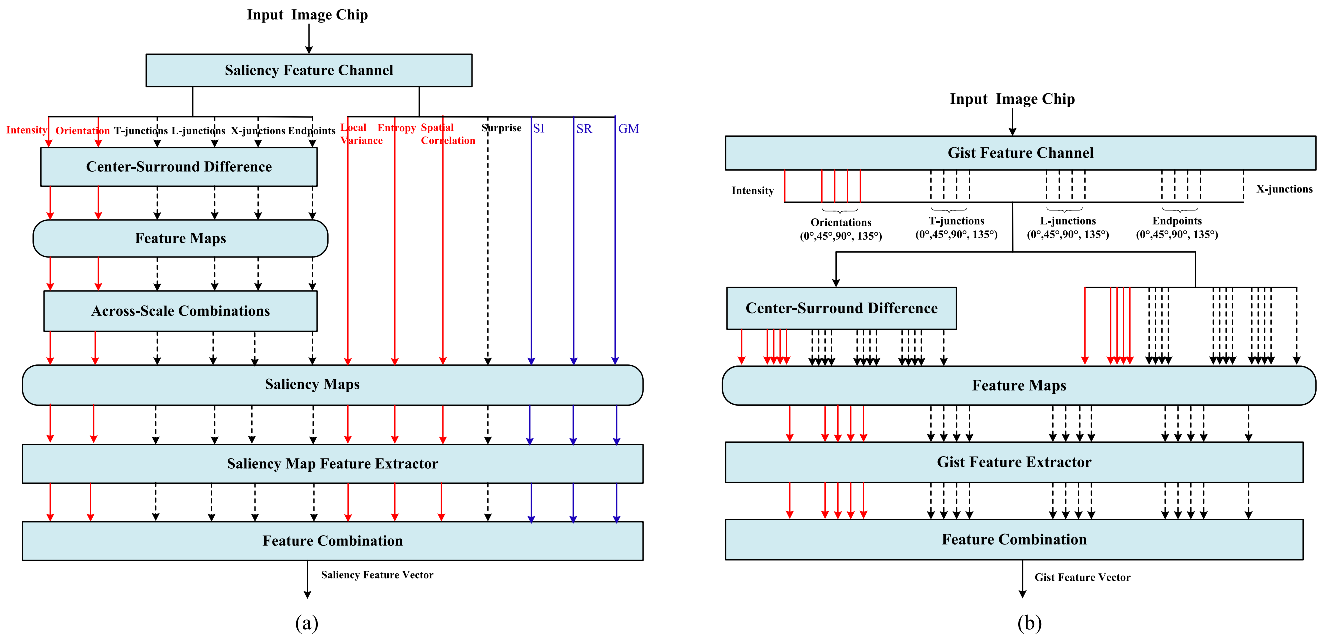


Fig. 5. Block diagram of feature extraction procedures of the MSG features for SAR images and the SG features for optical images. Here, the red solid lines and the blue solid lines correspond to the MSG features for SAR images, and the red solid lines and the black dotted lines correspond to the SG features for optical images, i.e., the MSG features exclude the features denoted by the black dotted lines from the SG features according to the characteristics of SAR images and add the new features denoted by the blue solid lines. (a) Saliency channel. (b) Gist channel.

images, Itti and Baldi [14] assumed D being Poisson distribution, $P(M)$ being Gamma distribution, and then $P(M|D)$ also being Gamma distribution. Here, the Gamma distribution is the conjugate prior for the Poisson distribution.

As mentioned above, for optical images, Itti *et al.* [10] assumed the image data being Poisson distribution and the prior distribution of the observer's beliefs being Gamma distribution. However, the Poisson distribution is obviously not suitable for SAR image data. Currently, some sophisticated statistical models [40], [41], such as Gamma, Weibull, G^0, K, and generalized Gamma distributions, are used to describe the high-resolution SAR images. If we employ these sophisticated distributions to model the SAR image data, the prior distribution of the belief model should be reassumed and the posterior distribution of the belief model should be rederivation. It is not easy to assume the appropriate prior distribution of the belief model when the data likelihood is sophisticated, and the appropriate prior distribution needs to be further researched. Therefore, we also exclude the surprise subchannel from the original SG features in our proposed MSG features.

B. Modified Saliency Features

The block diagram of feature extraction procedure for the modified saliency features is shown in Fig. 5(a), denoted by the red solid lines and blue solid lines. In this diagram, an image chip is input into the saliency channel and analyzed along multiple low-level feature subchannels. Eight-feature subchannels are adopted here: intensity, orientation combined with filters from 0°, 45°, 90°, and 135° (the details about combination will be introduced in the following), local variance, entropy, spatial correlation, SI, SR, and GM. Here, the SI, SR, and GM are the new added subchannels, and the other subchannels are from the original SG features. Then, eight saliency maps are generated from the eight subchannels, respectively. To decrease the dimensionality, a saliency map feature extractor is applied to transform each saliency map into a 4-D vector.

We give the details about each subchannel and the saliency map feature extractor in the following.

1) Intensity Subchannel: This subchannel generates an intensity saliency map in the same way as what the proposed target detection method does. As introduced in Section II-A, the saliency map is obtained by the across-scale combination of the feature maps, and each feature map is generated from the center-surround difference operation via the scale selection.

2) Orientation Subchannel: The orientation pyramid of an SAR image chip with the orientation filter of θ_k can be obtained as follows:

$$F_o(\sigma, k) = \mathrm{Gabor}(I_\sigma, \theta_k). \tag{10}$$

Here, $\mathrm{Gabor}(\cdot)$ is the Gabor filter, I_σ is the image chip of scale σ in the image Gaussian pyramid shown in Fig. 3, θ_k is the orientation of the Gabor filter (in this paper, $\theta_k = 0°, 45°, 90°, 135°$), and $F_o(\sigma, k)$ is the result of I_σ filtered by the Gabor filter with θ_k. Therefore, we can obtain k orientation pyramids of the SAR image chip. Then, for each orientation pyramid, a saliency map $I_s(\theta_k)$ is obtained in the same way as what the proposed target detection method does in Section II-A for each θ_k. Finally, the orientation saliency map is formed as

$$I_{s,o} = \sum_{k=1}^{4} I_s(\theta_k). \tag{11}$$

3) Local Variance Subchannel: The input SAR image chip is first divided into some 16×16 nonoverlapping image patches. For each 16×16 SAR image patch of the input SAR image chip, the variance can be calculated as

$$F_V = \left(\frac{\sum_j (I_{\text{patch}}(j) - m(I_{\text{patch}}))^2}{S_T - 1} \right)^{\frac{1}{2}} \tag{12}$$

where $I_{\text{patch}}(j)$ is the intensity of the jth pixel in the 16×16 patch, $m(I_{\text{patch}})$ is the mean intensity value over this image patch, and $S_T = 16 \times 16$ is the total pixel number of the image patch. In this way, we can obtain a local variance saliency map. The local variance not only can measure the local fluctuation of the pixel intensities of the image chip, but also can capture some spatial information of the target or clutter.

4) Entropy Subchannel: The entropy value of each 16×16 image patch can be computed as

$$F_E = -\sum_{z=0}^{255} p(z) \times \log(p(z)), \quad p(z) = \frac{N(z)}{N_{\text{total}}} \tag{13}$$

where $p(z)$ is the probability of the intensity value z in I_{patch}, $N(z)$ represents the number of pixels with the intensity value the same as z in I_{patch}, and N_{total} is the total number of pixels in I_{patch}. Note that the intensity values of I_{patch} are normalized to 0–255 in advance. In particular, when $p(z) = 0$, we let $p(z) \times \log(p(z)) = 0$. Then, similar to the local variance subchannel, we obtain an entropy saliency map. Entropy indicates the probability distribution of the intensities of the SAR image patches.

5) Spatial Correlation Subchannel: For two random variables X and Y, their correlation is expressed as

$$\begin{aligned} \rho_{X,Y} &= \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} \\ &= \frac{E(XY) - E(X) \times E(Y)}{\sqrt{E(X^2) - E^2(X)}\sqrt{E(Y^2) - E^2(Y)}}. \end{aligned} \tag{14}$$

Here, the spatial correlation is computed between a local 16×16 patch and its neighborhood patches. In this paper, we select eight neighborhoods, and the mean value of the eight correlations is adopted as the spatial correlation of this patch. This subchannel generates a spatial correlation saliency map. In the SAR images, the targets usually present lower spatial correlation than the background clutter.

6) SI Subchannel: The input SAR image chip is also first divided into some 16×16 nonoverlapping local image patches. Then, for each SAR image patch, the SI [13] saliency can be computed as

$$F_I = -\log p(I_{\text{patch}}) \tag{15}$$

$$p(I_{\text{patch}}) = \prod_j p_{\text{chip}}(z_j) \tag{16}$$

$$p_{\text{chip}}(z_j) = \frac{N(z_j)}{N_{\text{chip}}} \tag{17}$$

where $p(I_{\text{patch}})$ is the joint probability of the pixels in the local image patch, $p_{\text{chip}}(z_j)$ is the probability of the jth pixel in the image chip, $N(z_j)$ is the number of pixels with the intensity value the same as z_j in the image chip, and N_{chip} is the total number of pixels in the image chip. Then, with calculating the SI of each image patch, we can obtain an SI saliency map. The target usually has higher SI than that of the clutter, since the target pixels are only a small part of the SAR image chip.

7) SR Subchannel: The SR [15] saliency map of an SAR image chip is calculated as

$$F_{SR} = G(x) * F^{-1}[\exp(R(f) + P(f))]^2 \tag{18}$$

$$R(f) = \log(A(f)) - h(f) * \log(A(f)) \tag{19}$$

$$A(f) = \text{Re}(F(I_{\text{chip}})) \tag{20}$$

$$P(f) = \text{Im}(F(I_{\text{chip}})) \tag{21}$$

where $A(f)$ represents the amplitude spectral of the image chip I_{chip}, $P(f)$ is the phase spectral of I_{chip}, $R(f)$ is the SR, $F(\cdot)$ is the Fourier transform, $F^{-1}(\cdot)$ is the inverse Fourier transform, $\text{Re}(\cdot)$ represents the real part of the complex number, $\text{Im}(\cdot)$ denotes the imaginary part of the complex number, $G(x)$ is the Gaussian filter, and $h(f)$ is the mean filter. The SR saliency can highlight the targets while suppressing the background clutter in the whole image chip quickly with the fast Fourier transform.

8) GM Subchannel: Here, the ratio of exponentially weighted averages (ROEWA) [42] rather than the difference is used to compute the GM, which is more robust to multiplicative noise for SAR images. The ROEWA-GM of a given pixel point (i, j) in an SAR image chip is calculated as follows:

$$G(i,j) = \sqrt{(G_x(i,j))^2 + (G_y(i,j))^2} \tag{22}$$

$$\begin{aligned} G_x(i,j) &= \log\left(\frac{M_{x,1}(i,j)}{M_{x,2}(i,j)}\right) \\ G_y(i,j) &= \log\left(\frac{M_{y,1}(i,j)}{M_{y,2}(i,j)}\right) \end{aligned} \tag{23}$$

$$M_{x,1}(i,j) = \sum_{x=-l}^{-1} \sum_{y=-\lfloor l/2 \rfloor}^{\lfloor l/2 \rfloor} (i+x, j+y) e^{-\frac{|x|+|y|}{\alpha}} \tag{24}$$

$$M_{x,2}(i,j) = \sum_{x=1}^{l} \sum_{y=-\lfloor l/2 \rfloor}^{\lfloor l/2 \rfloor} (i+x, j+y) e^{-\frac{|x|+|y|}{\alpha}} \tag{25}$$

$$M_{y,1}(i,j) = \sum_{x=-\lfloor l/2 \rfloor}^{\lfloor l/2 \rfloor} \sum_{y=-l}^{-1} (i+x, j+y) e^{-\frac{|x|+|y|}{\alpha}} \tag{26}$$

$$M_{y,2}(i,j) = \sum_{x=-\lfloor l/2 \rfloor}^{\lfloor l/2 \rfloor} \sum_{y=1}^{l} (i+x, j+y) e^{-\frac{|x|+|y|}{\alpha}} \tag{27}$$

where $G(i, j)$ is the ROEWA-GM of the pixel point (i, j), $G_x(i, j)$ and $G_y(i, j)$ are the horizontal and vertical gradients, respectively, $M_{x,1}(i, j)$ and $M_{x,2}(i, j)$ are the exponential weighted means of the local patches on the left and right of the point (i, j), respectively, $M_{y,1}(i, j)$ and $M_{y,2}(i, j)$ are the exponential weighted means of the local patches above and below the point (i, j), respectively, $\lfloor \cdot \rfloor$ is the operation of rounding down, l is the length of the local patch, and α is the exponential weight parameter. Here, when l is set to 3–7, the proposed discrimination features usually can obtain satisfactory results, and in this paper, l is set to 3; α is set to 2, same as the corresponding setting used in [42].

The ROEWA-GM of the target is usually much stronger than those of the clutter, and the outlines of the targets reflected by the GM are usually much different from those of the clutter.

9) Saliency Map Feature Extractor: To decrease the dimensionality, a saliency map feature extractor is applied to transform each saliency map into a 4-D vector with mean, variance, number of local maxima points, and average distance between locations of local maxima [11]. All those feature vectors from the five saliency feature subchannels are combined into a vector called "modified saliency features."

C. Modified Gist Features

The block diagram of feature extraction procedure for the modified gist feature for SAR target discrimination in this paper is shown in Fig. 5(b), denoted by the red solid lines.

1) Intensity and Orientation Subchannels: Here, five different feature subchannels are adopted: intensity subchannel and four orientation subchannels from 0°, 45°, 90°, and 135°, which are not combined into one subchannel as the modified saliency feature extraction does. The operations of the five subchannels are similar to those of the intensity and orientation subchannels for the modified saliency features, except that the inputs of the gist feature extractor are "feature maps" instead of "saliency maps."

The modified gist feature extraction procedure uses both the center-surround and raw (before center surround) feature maps. The center-surround feature maps are computed within the pyramid corresponding to each subchannel, where the scales for them are selected in the same way as what the proposed target detection method does in Section II-A. The raw feature maps are generated from the pyramid corresponding to each subchannel with scales ranging from 0 to n (in this paper, n is set to 5); therefore, we can obtain $n+1$ raw feature maps for each subchannel.

2) Gist Feature Extractor: Here, the mean value is adopted to represent each gist feature map, which can describe an SAR image chip's overall information. Accordingly, the purpose of the modified gist features is to summarize the global information of the scene, rather than to highlight the local saliency objects of the scene as the modified saliency features do.

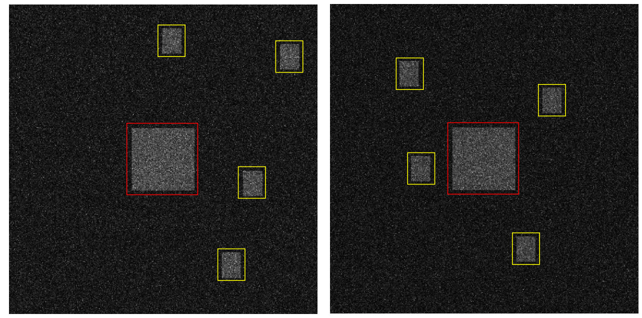
IV. Experimental Results and Analysis

The effectiveness of the proposed detection method is verified by experiment 1 using both the synthetic images and real SAR images. Then, the effectiveness of the proposed discrimination method is verified by experiment 2 using real SAR image chips extracted by the proposed target detection method.

The following two experiments are conducted using a PC with Intel Core i5-4590 CPU of 3.3 GHz and memory of 8 GB. The program codes are written in MATLAB R2012a.

A. Experiment 1

In experiment 1, we compare the proposed detection method with the Itti's method [10], the SVD method [19], and the two-parameter CFAR method [5], [6] on both the synthetic images and real SAR images. Here, the two-parameter CFAR method is the classical detection method for SAR images, while the other two detection methods are the visual attention-based detectors. The receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) at the pixel level are employed to evaluate the detection performance of different detection methods. Here, the probability of detection in the ROC curve is defined as the ratio of the number of detected target pixels to that of total target pixels, and the probability of false alarm is defined as the ratio of the number of detected clutter pixels to that of total clutter pixels.



(a) (b)

Fig. 6. Two examples of synthetic images. (a) and (b) Synthetic images with sizes of 500 × 500 pixels. The targets of interest with sizes of 41 × 31 pixels are marked by the yellow rectangles, while the objects not of interest with sizes of 101 × 101 pixels are marked by the red rectangles.

1) Target Detection of Synthetic Images: The synthetic images are generated in the similar way to those in [8] and [34]. Then, Monte Carlo simulations repeat 1000 times. In each simulation, four targets are randomly distributed in one synthetic image and the intensities of the multiplicative speckle noises are also different. Fig. 6 shows two examples of the synthetic images. Here, the objects marked by the yellow rectangles are the targets of interest with sizes of 41 × 31 pixels; while the objects marked by the red rectangles with sizes of 101 × 101 pixels, which are larger than the targets of interest, and the background is the clutter.

Parameters Setting: For the proposed detection method, the center scale is set to $c = \{2\}$, and the surround scales are set to $s = \{4, 5, 6\}$. Note that here the best surround scale is round$(\log_2^{31}) = 5$, and scales 4 and 6 close to scale 5 are also selected as surround scales. For both the Itti's method and SVD method, the center scales are set to $c = \{2, 3, 4\}$, and the surround scales are set to $s = c + \delta$ with $\delta = \{3, 4\}$, same as the corresponding setting used in [10] and [19]. For the two-parameter CFAR target detection method, the target window is 1×1 pixels, the guard window is 105×105 pixels, and the background window is 107×107 pixels.

Fig. 7 shows different detection methods' average ROC curves of 1000 synthetic images. The closer to the top left the ROC curve, the better is its performance. It can be seen from Fig. 7 that the ROC curve of the proposed detection method is better than those of the other detection methods (i.e., the Itti's method, SVD method, and two-parameter CFAR). Furthermore, Table I gives different detection methods'

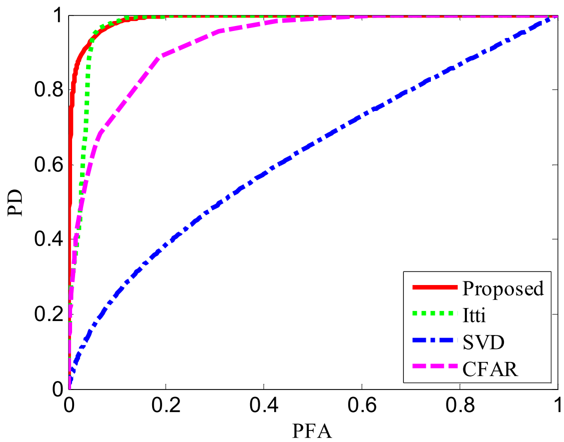


Fig. 7. Different detection methods' average ROC curves of 1000 synthetic images. Here, "Proposed" is the detection method proposed in this paper, "Itti" is the Itti's method [10], "SVD" is the SVD method [19], and "CFAR" is the two-parameter CFAR method [5], [6].

TABLE I
DIFFERENT DETECTION METHODS' AVERAGE AUCS OF 1000 SYNTHETIC IMAGES

	Proposed	Itti	SVD	CFAR
AUC	**0.9888**	0.9742	0.6224	0.9264

(a)

(b)

Fig. 8. Example of the miniSAR data set. (a) SAR image from miniSAR data set. (b) Optical image from Google Earth whose scene corresponds to (a).

average AUCs of 1000 synthetic images. The larger the AUC, the better is its performance. As shown in Table I, the proposed detection method has the largest AUC.

2) Target Detection of Real SAR Images: The miniSAR real SAR image data set is acquired by Sandia National Laboratories of America.[1] The scenes of the SAR images in this data set are rather complex, where there are vehicles, helicopters, buildings, roads, grasslands, trees, streetlights, and so on. Fig. 8 gives an example of SAR image from the miniSAR data set, and we also show the optical image from Google Earth corresponding to the scene of the SAR image (note that the optical scene is not acquired the same time as the SAR image, and of course they are not completely corresponding). We select two SAR images that have vehicles from the miniSAR data set to illustrate the effectiveness of the proposed detection method, and their resolutions are 0.1 m $\times$ 0.1 m. The two SAR images are shown in Fig. 9(a) and (b), of which the sizes are 1638×2510 pixels, the vertical axes represent the range dimensions, and the horizontal axes represent the azimuth dimensions. Here, the vehicles are the targets of interest, and the others, such as buildings, trees, and grasslands, are treated as clutter. The ground truths of vehicle targets of Fig. 9(a) and (b) are given in Fig. 9(c) and (d), respectively, which are marked by white color manually by visual inspection. For convenience, we refer to Fig. 9(a) as Image I and Fig. 9(b) as Image II.

Parameters Setting: For the miniSAR real data set, we are interested in the civil vehicles, especially including hatchbacks, notchbacks, sport utility vehicles, and mini vans. In general, the widths of these civil vehicles are about 2–3 m and the lengths of these civil vehicles are about 4–5 m. Then, the prior maximum width and length of the civil vehicles are about 3 and 5 m, respectively. According to the resolution (0.1 m $\times$ 0.1 m) of the SAR images in Fig. 9(a) and (b), the vehicle targets have sizes of about 50 $\times$ 30 pixels. For the proposed detection method, the center scale is set to $c = \{2\}$, and the surround scales are set to $s = \{4, 5, 6\}$. Note that here the best surround scale is $\text{round}(\log_2^{30}) = 5$, and scales 4 and 6 close to scale 5 are also selected as surround scales. For both the Itti's method and SVD method, the center scales are set to $c = \{2, 3, 4\}$, and the surround scales are set to $s = c + \delta$ with $\delta = \{3, 4\}$, same as the corresponding setting used in [10] and [19]. For the two-parameter CFAR, the target window is 1×1 pixels, the guard window is 97×97 pixels, and the background window is 99×99 pixels.

To better compare different detection methods, in this paper, we define the saliency map of the two-parameter CFAR as

$$\frac{X_t - \hat{\mu}_c}{\hat{\sigma}_c} \Rightarrow \text{Saliency map} \tag{28}$$

where X_t is the intensity of the current pixel under test, $\hat{\mu}_c$ is the estimated intensity mean of the surrounding pixels around the current pixel and within the background window, and $\hat{\sigma}_c$ is the estimated intensity standard deviation of the surrounding pixels around the current pixel and within the background window. Fig. 10 shows the saliency maps of Image I and Image II obtained by different detection methods. Fig. 10(c) and (d) shows the saliency maps obtained by the

[1] http://www.sandia.gov/radar/imagery

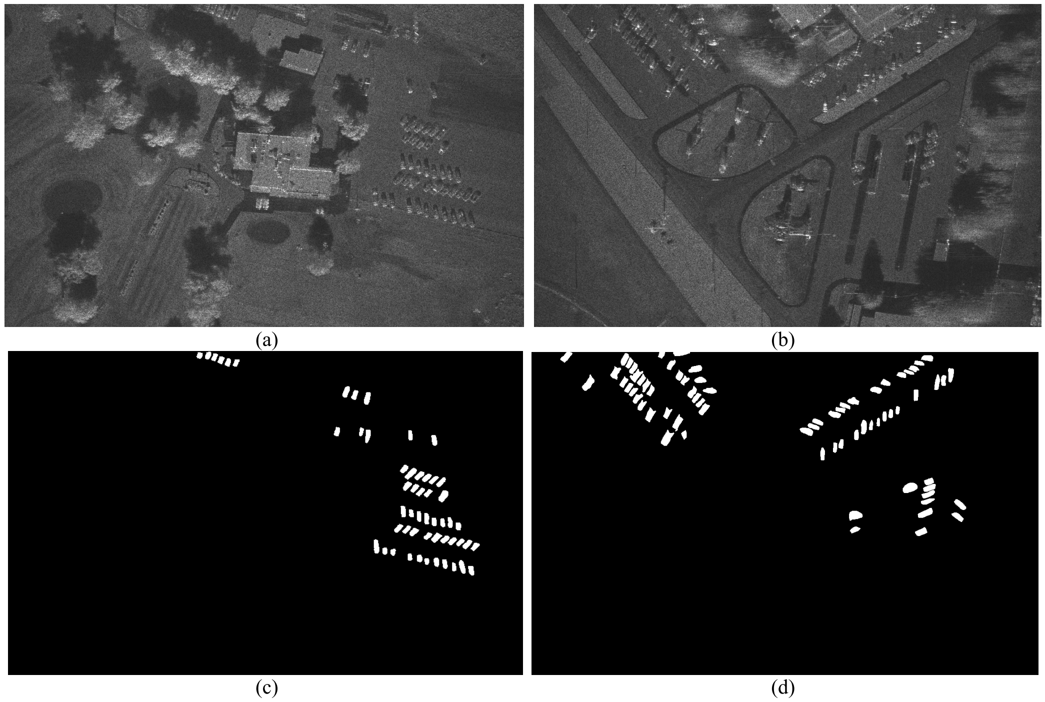


Fig. 9. Real SAR images and their ground truths used to verify the detection performance of different detection methods. (a) Image I used in experiment 1. (b) Image II used in experiment 1. (c) and (d) Ground truths of the vehicle targets of (a) and (b), respectively. The ground truths of vehicle target regions in (c) and (d) are marked by white color manually by visual inspection.

TABLE II
AUCs OF DIFFERENT DETECTION METHODS

Image	Proposed	Itti	SVD	CFAR
Image I	**0.7477**	0.6524	0.6703	0.6685
Image II	**0.7822**	0.6617	0.6769	0.6807

TABLE III
TIME COST OF OBTAINING SALIENCY MAPS VIA DIFFERENT DETECTION METHODS (SECOND)

Image	Proposed	Itti	SVD	CFAR
Image I	**0.45**	0.58	7.25	284.64
Image II	**0.46**	0.59	7.03	290.29

Itti's detection method, from which we can see that not only the vehicle targets but also numerous building and tree clutter false alarms are highlighted. As shown in Fig. 10(e) and (f), the saliency maps obtained by the SVD detection method can only highlight a part of target pixels and the structures of the targets are incomplete. It can be seen from Fig. 10(g) and (h) that the saliency maps obtained by the two-parameter CFAR are similar to the original SAR images shown in Fig. 9(a) and (b) and the vehicle targets are almost not highlighted. On the contrary, as shown in Fig. 10(a) and (b), the saliency maps obtained by the proposed detection method not only highlight the vehicle targets but also suppress most clutter smoothly.

To quantitatively evaluate the detection performance of different detection methods at the pixel level, the ROC curves of different detection methods are shown in Fig. 11 and the corresponding AUCs are shown in Table II. It can be seen from Fig. 11 and Table II that the proposed detection method has the best detection performance among all the detection methods. In addition, Table III gives the time cost of obtaining the saliency maps via different detection methods, from which we can see that the proposed detection method has the smallest time cost among all the detection methods.

The reasons for the proposed detection method not only obtaining the best detection performance but also having the smallest time cost among all the detection methods can be summarized as follows.

First, for the proposed detection method, we select the task-dependent scales based on the size prior information of the vehicle targets for constructing the saliency map to highlight the vehicle targets and suppress the background clutter, such as trees and buildings, in the SAR image. Therefore, the targets will be detected from the complex background clutter with high priority. However, both the Itti's method and the SVD method used the fixed scales to construct the saliency maps for different detection task; thus, the targets of interest cannot be guaranteed to be more salient than the background clutter. Moreover, since the scenes of Image I and Image II are very complex, the clutter statistical model used in the two-parameter CFAR cannot match with the actual clutter distribution, which inevitably results in a large number of false alarms or missed targets.

Second, the proposed detection method uses the time-saving center-surround mechanism (i.e., the difference between the two images with different scales) to construct the saliency

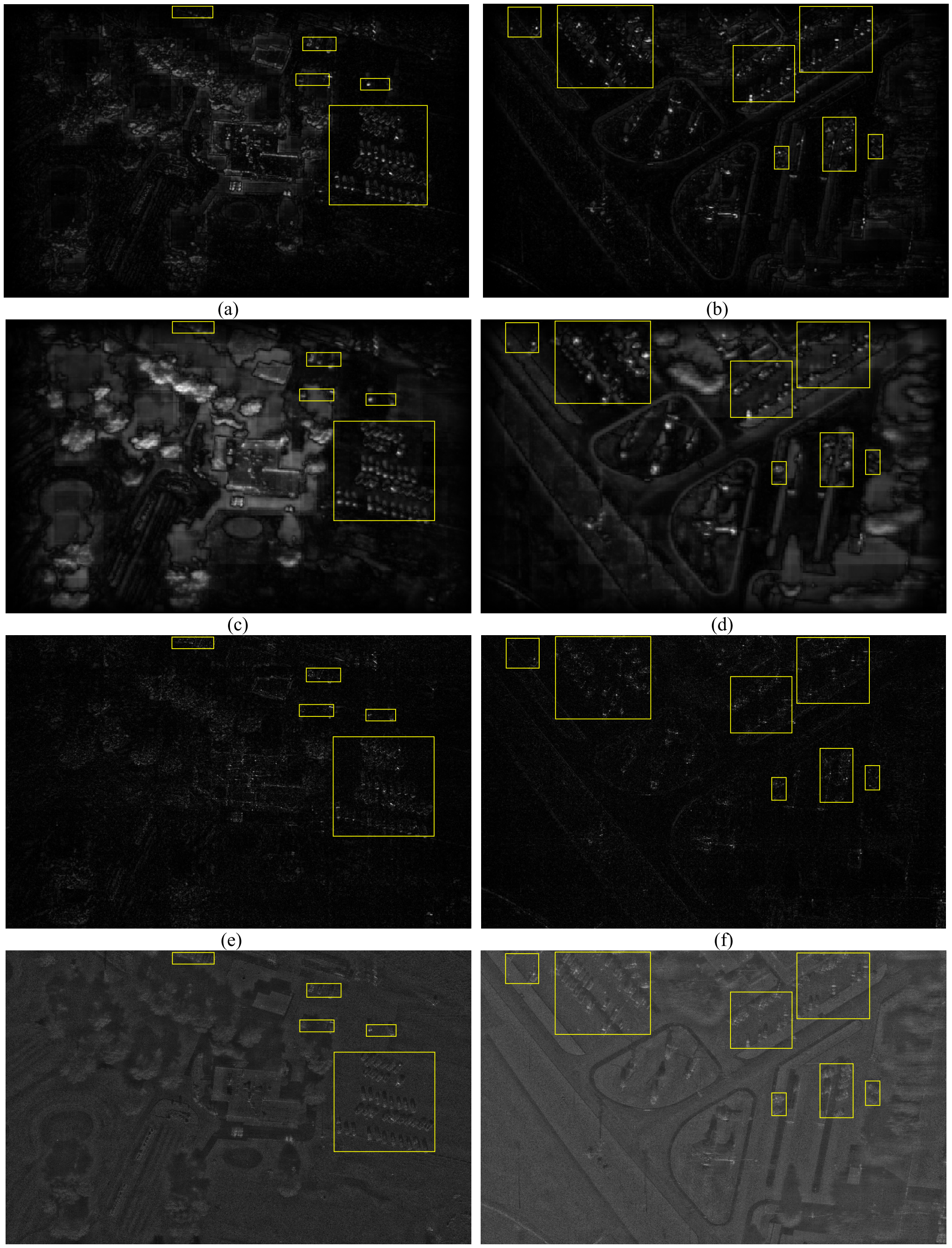
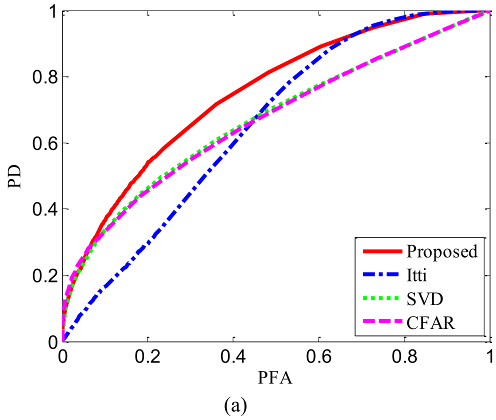


Fig. 10. Saliency maps of the SAR images obtained by different detection methods. Saliency maps of Image I obtained by (a) proposed method, (c) Itti's method, (e) SVD method, and (g) two-parameter CFAR. Saliency maps of Image II obtained by (b) proposed method, (d) Itti's method, (f) SVD method, and (h) two-parameter CFAR. Here, the vehicle target regions are marked by the yellow rectangles. The saliency map of the two-parameter CFAR is defined as (28) in this paper for comparison.

map, instead of using the time-consuming sliding window mechanism (i.e., estimating the model parameter of the clutter around each pixel in the whole image) as the two-parameter CFAR does. In addition, we only select several scales related to the detection task, rather than using many scales as the Itti's method and the SVD method did.

To further illustrate the advantages of the proposed detection method, we apply the postprocessing operations, including morphological filtering, clustering, and false-alarm region removing, i.e., steps 4–6 shown in Fig. 2, to the binary images obtained by different detection methods, and then compare their detection performance with postprocessing. In the following, we adopt Image I to validate the effectiveness of the postprocessing. Here, the binary images of the proposed detection method, Itti, and SVD are obtained via the operation of "shifts of the FOA" in their saliency map, and the binary image of CFAR is obtained by thresholding the saliency map in (28).



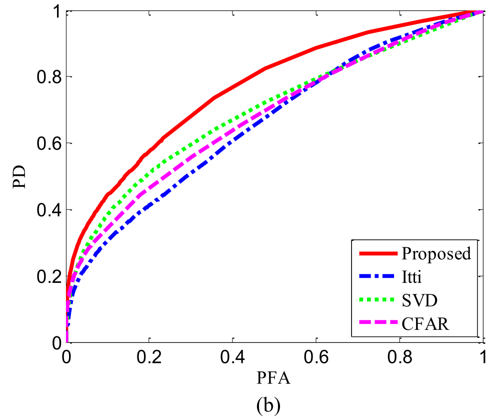


Fig. 11. ROC curves of different detection methods. (a) Image I. (b) Image II.

TABLE IV

DETECTION RESULTS OF DIFFERENT DETECTION METHODS WITH POSTPROCESSING (THE NUMBER OF ALL THE TARGETS IS 54)

Method	N_{de}	N_{mi}	N_{fa}	T_{loss}
CFAR	40	14	44	72
Itti	8	46	**0**	92
SVD	39	15	269	299
Proposed	**43**	**11**	29	**51**

Fig. 12 shows the detection binary images and their corresponding postprocessing results of the CFAR, Itti, SVD, and proposed detection method, respectively. As shown in Fig. 12, compared with the other detection methods with postprocessing, the proposed detection method with postprocessing can not only detect most targets but also generate few false alarms.

Table IV quantitatively evaluates the results of different detection methods with postprocessing at the target level, using the number of detected targets (N_{de}), number of missed targets (N_{mi}), number of false alarms (N_{fa}), and detection loss (T_{loss}). Here, T_{loss} is defined as

$$T_{\mathrm{loss}} = N_{\mathrm{fa}} + \beta N_{\mathrm{mi}} \tag{29}$$

where β specifies a relative importance of N_{fa} versus N_{mi}. In this experiment, β is set to 2, which can avoid missing too many targets.

As shown in Table IV, the proposed detection method outperforms the CFAR method, Itti method, and SVD method in terms of N_{de}, N_{mi}, and T_{loss} has fewer N_{fa} than that of the CFAR method and SVD method, and has more N_{fa} than that of the Itti method. Although the Itti method has fewer false alarms, it misses much more targets than the proposed detection method.

Fig. 13 shows the candidate target chips of Images I and II finally extracted by the proposed detection method. It can be seen from Fig. 13 that most vehicle targets are detected and only a few false alarm chips similar to the targets are generated. In the following, we will use the proposed discrimination method to discriminate the target chips from the clutter chips.

B. Experiment 2

In experiment 2, we select eight SAR images that have vehicles from the miniSAR data set for our experiments. Then, the candidate target chips (these chips are called as miniSAR chip data set) extracted by the proposed detection method from the eight SAR images are sent into the discrimination stage. There are 590 chips, including 286 target chips and 304 clutter chips.

Examples of target chips and clutter chips are shown in Fig. 14, from which we can see that the extracted candidate target chips mainly contain the vehicle targets, buildings, and other complex clutter, making it difficult to discriminate the targets from clutter. In the following, we compare the proposed MSG features with the old Lincoln features [4], [24], new Lincoln features [25], Gao's features [21], [28], Bhanu's features [26], [27], SG features [11], and JS features (i.e., the features which are excluded from the original SG features).

1) Analysis on Feature Separability: The ratio of between-class distance to within-class distance (RBTW) [29] is adopted to compare the linear separability of different discrimination features. We execute five runs for the linear separability comparison experiment. In each run, we randomly select 200 target chips and 200 clutter chips from the miniSAR chip data set. It can be seen from Fig. 15 that the RBTW of the MSG features is much larger than those of the other discrimination features under each random data division. Therefore, the MSG features can separate the targets from the clutter much more easily than the other features.

2) Analysis on Discrimination Results: First, we execute 100 runs for the discrimination experiment, selecting the training and test data randomly for each run. We select four SAR images as training set (including 138 target chips and 158 clutter chips) and four SAR images as test set (including 148 target chips and 146 clutter chips). In each run, 100 target chips and 100 clutter chips are selected randomly from training set for training, and 100 target chips and 100 clutter chips are selected randomly from test set for testing.

Table V compares the discrimination performance of different discrimination features using the linear SVM, where the probability of detection Pd, the probability of false alarm Pf, and the probability of correct discrimination Pc at the chip

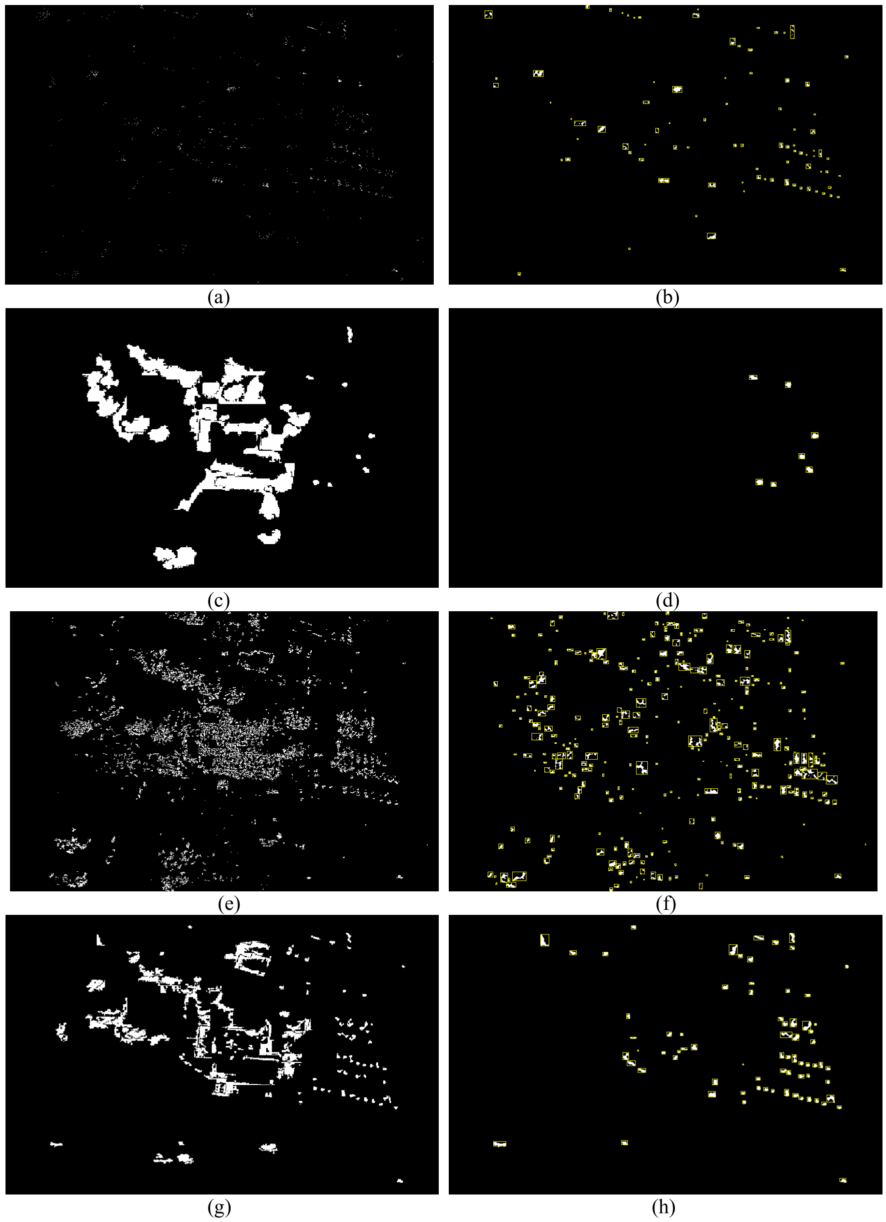


Fig. 12. Binary images obtained by different detection methods and their postprocessing results of Image I. Binary images obtained by (a) CFAR, (c) Itti, (e) SVD, and (g) proposed detection method. Here, both the false alarm rate P_{fa} for CFAR and the target confidence levels α for the Itti, SVD, and proposed detection method are set to be 0.4%. (b), (d), (f), and (h) Final detection results with postprocessing, i.e., morphological filtering, clustering, and false-alarm region removing. Note that here the yellow rectangles are the smallest rectangle containing each cluster.

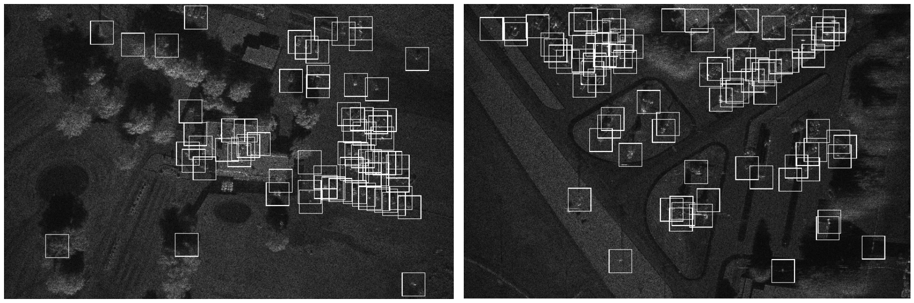
level are calculated as follows:

$$Pd = \frac{\text{Number of target chips detected as target}}{\text{Total number of target chips}} \tag{30}$$

$$Pf = \frac{\text{Number of clutter chips detected as target}}{\text{Total number of clutter chips}} \tag{31}$$

$$Pc = \frac{Pd + (1 - Pf)}{2}. \tag{32}$$

As shown in Table V, the proposed MSG features have much higher Pd and Pc than those of the other discrimination features, and have lower Pf than that of the other discrimination features except for the Gao's features. Although the Gao's features have the lowest Pf, they have much lower Pd and Pc than those of the MSG features. Table VI shows a confusion matrix of the discrimination result of the MSG features in one random run. Fig. 16 shows the average ROC curves and the corresponding AUCs of different discrimination features for the 100 runs using the linear SVM discriminator. As shown in Fig. 16, the average ROC curve of the MSG features is the best among all discrimination features, and the corresponding AUC of the MSG features is also much larger than those of the other discrimination features. In addition, the Pcs of different discrimination features with different numbers of training samples are shown in Fig. 17, from which we can see that the discrimination performance of the MSG features is better than those of the other discrimination features with different numbers of training samples. The reasons for the MSG features obtaining the best discrimination performance



(a) (b)

Fig. 13. Candidate target chips extracted by the proposed detection method. (a) Image I. (b) Image II. Here, the white rectangles represent the extracted chips.

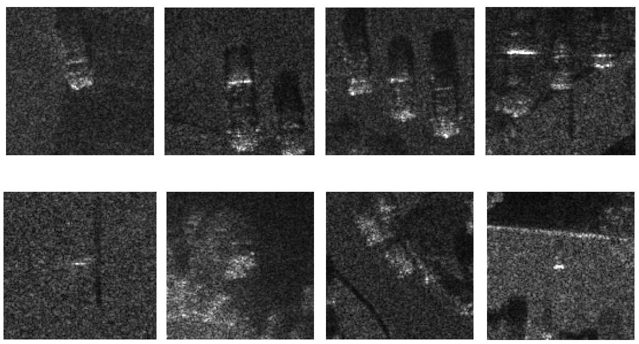


Fig. 14. Examples of the candidate vehicle target chips acquired by the proposed target detection method. First row: real vehicle targets. Second row: some false detections coming from the clutter. Each target chip contains one or more vehicle targets with some ground clutter, and the clutter chips contain the tree, building, or the grassland clutter.

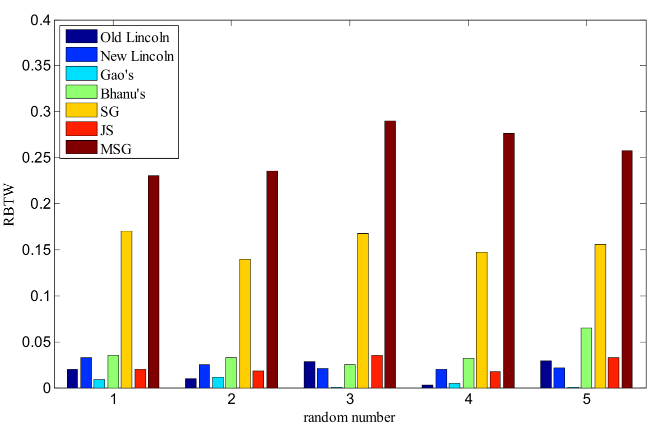


Fig. 15. RBTW values of different discrimination features in the five runs.

TABLE V

COMPARISON OF THE DISCRIMINATION PERFORMANCE AMONG DIFFERENT DISCRIMINATION FEATURES, ALL USING THE LINEAR SVM DISCRIMINATOR

Discrimination features	*Pd*(%)	*Pf*(%)	*Pc*(%)
Old Lincoln	76.29 ± 4.33	31.65 ± 3.76	72.32 ± 2.78
New Lincoln	41.54 ± 6.85	24.72 ± 6.32	58.41 ± 2.52
Gao's	20.30 ± 15.98	**6.53** ± 16.79	56.88 ± 1.67
Bhanu's	61.02 ± 12.92	34.80 ± 13.97	63.11 ± 3.36
SG	80.19 ± 3.71	21.99 ± 3.90	79.01 ± 2.55
JS	71.60 ± 4.68	40.91 ± 7.92	65.35 ± 3.83
MSG	**83.94** ± 2.83	21.32 ± 3.71	**81.31** ± 2.21

TABLE VI

CONFUSION MATRIX OF THE DISCRIMINATION RESULT OF THE MSG FEATURES IN ONE RANDOM RUN

	actual positive	actual negative
predicted positive	0.89	0.23
predicted negative	0.11	0.77

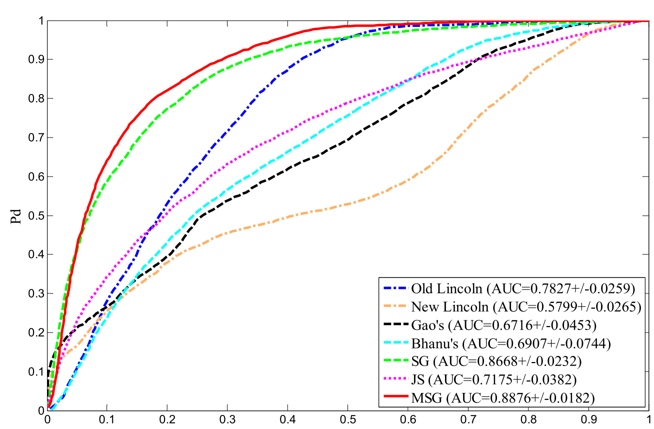


Fig. 16. Average ROC curves and the corresponding AUCs of different discrimination features for 100 runs using the linear SVM discriminator.

among all the discrimination features can be summarized as follows.

First, since there may be single target, multiple targets, or partial targets with complex clutter in each SAR image chip, it is hard to acquire accurate target-shaped blob via segmentation. Accordingly, some classical discrimination features, such as the old Lincoln features and the Bhanu's features, which are extracted based on target segmentation, have unsatisfactory discrimination performance. On the contrary, the MSG features, including the SG feature channels, are complementary to each other and can provide a complete description of the extracted SAR image chips without segmentation.

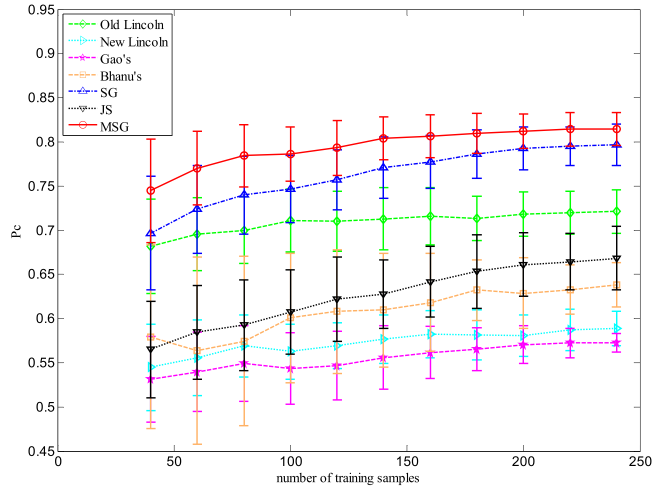


Fig. 17. Pcs of different discrimination features with different numbers of training samples (half from targets and the other half from clutter). The error bars are calculated from 100 runs with randomly selecting the samples from the training set.

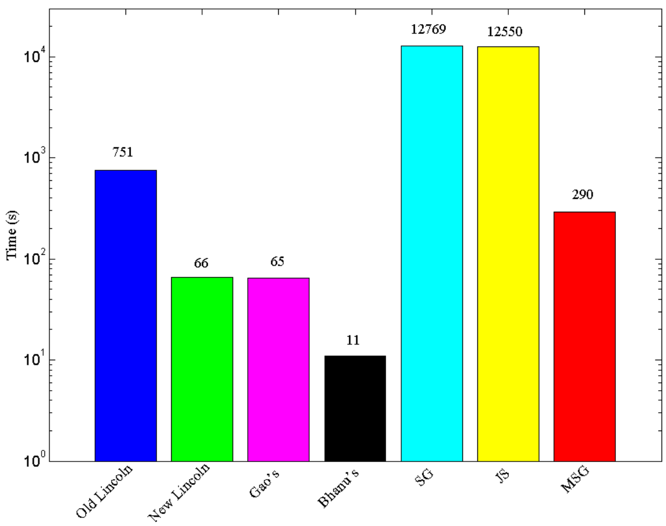


Fig. 18. Comparison of the time costs among the feature extraction procedures of different discrimination features.

Second, as shown in the above experiments, the JS features have unsatisfactory discrimination performance which indicates that the JS features are not suitable for target discrimination in SAR images. Owing to excluding the JS features, the MSG features are more suitable for the application in SAR images.

Finally, we add three new subchannels which are fit for SAR target discrimination into the MSG features. The three new subchannels include the SI, SR, and GM subchannels, and the targets and clutter usually present different characteristics in these new subchannels.

3) Analysis on Time Cost for Feature Extraction: The comparison of the feature extraction time costs among different discrimination features is shown in Fig. 18. Obviously, it can be seen from Fig. 18 that the time cost of the MSG features is much less than that of the old Lincoln features, SG features, and JS features, and a little larger than those of the new Lincoln features, Gao's features, and Bhanu's features. The reasons for the MSG features having a relative low time cost of the feature extraction can be summarized as follows.

First, owing to excluding the JS features, the MSG features are more suitable for real-time practical application in SAR images. As shown in Fig. 18, the extraction of the JS features is time consuming. The extraction procedure of the junction features is very complex and we take the T-junctions as an example for illustration. There are nine images with different scales in the T-junction pyramid, and each image is obtained via eight binary masks filtering the corresponding images with the same scale in the four orientation pyramids. Therefore, the construction of the T-junction pyramid requires filtering $9 \times 8 \times 4 = 288$ times, where "9" is the number of images in the pyramid, "8" is the number of binary filter masks, and "4" is the number of orientation pyramids. However, the construction of the intensity or orientation pyramid for MSG features only requires filtering 9 times. In addition, the surprise requires estimating the parameters of the prior and posterior distributions of each image patch in the feature maps in a sliding window manner, which is also time consuming.

Second, the extraction procedure of the MSG features only requires some simple filtering processing, e.g., Gaussian filtering, rather than using the complex operations, e.g., segmentation preprocessing and local CFAR operation in the old Lincoln features.

Finally, the three new added subchannels are not complex, and all of them can be extracted without costing too much time.

V. Conclusion

To overcome the drawbacks of the conventional methods of target detection and discrimination in high-resolution SAR images with complex scenes, in this paper, we propose a visual attention-based target detection and discrimination method. The major contributions of this paper are summarized as follows.

1) We select the task-dependent scales from the Gaussian pyramid of the original SAR image based on the size prior information of the targets to be detected, rather than using the fixed scales as Itti did.
2) We adopt the clustering method to connect several isolated FOA areas into a complete target region, instead of regarding every isolated FOA area as one complete target in the Itti's model.
3) We propose the MSG features for SAR target discrimination. The MSG features are more effective in dealing with the complex scenes in the SAR image chips than the classical discrimination features.

The experimental results based on the synthetic images and the miniSAR real data set demonstrate that the proposed target detection and discrimination method can detect and discriminate the targets from the clutter with high accuracy and fast computation speed.

In some applications, some new and complex classification methods, e.g., the deep learning-based target classification methods [34], [35], can achieve good classification performance. However, these methods require a large number of training data. For the target discrimination problem researched in this paper, the number of training data is very limited, which is not suitable for most of the new and complex classification methods. We have tried two deep networks, i.e., the stacked autoencoders [34] and the deep belief network [35], directly on our data set, and their probabilities of correct discrimination are not higher than 65%. In our future work, we will study on how to improve the discrimination performance of some new and complex classification methods for SAR target discrimination.

ACKNOWLEDGMENT

The authors would like to thank the editors and anonymous reviewers for their valuable suggestions and comments that greatly improved this paper.

REFERENCES

[1] L. M. Novak, G. J. Owirka, and C. M. Netishen, "Performance of a high-resolution polarimetric SAR automatic target recognition system," *Lincoln Lab. J.*, vol. 6, no. 1, pp. 11–24, 1993.

[2] L. M. Novak, S. D. Halversen, G. J. Owirka, and M. Hiett, "Effects of polarization and resolution on the performance of a SAR automatic target recognition system," *Lincoln Lab. J.*, vol. 8, no. 1, pp. 49–68, 1995.

[3] L. M. Novak, G. J. Owirka, W. S. Brower, and A. L. Weaver, "The automatic target-recognition system in SAIP," *Lincoln Lab. J.*, vol. 10, no. 2, pp. 187–202, 1997.

[4] D. E. Kreithen and L. M. Novak, "Discriminating targets from clutter," *Lincoln Lab. J.*, vol. 6, no. 1, pp. 25–51, 1993.

[5] G. B. Goldstein, "False-alarm regulation in log-normal and Weibull clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-9, no. 1, pp. 84–92, Jan. 1973.

[6] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.

[7] G. Gao, L. Liu, L. Zhao, G. Shi, and G. Kuang, "An adaptive and fast CFAR algorithm based on automatic censoring for target detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1685–1697, Jun. 2009.

[8] D. Tao, S. N. Anfinsen, and C. Brekke, "Robust CFAR detector based on truncated statistics in multiple-target situations," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 117–134, Jan. 2016.

[9] M. Greenspan, L. Pham, and N. Tardella, "Development and evaluation of a real time SAR ATR system," in *Proc. IEEE RADAR*, May 1998, pp. 38–43.

[10] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 11, pp. 1254–1259, Nov. 1998.

[11] Z. Li and L. Itti, "Saliency and gist features for target detection in satellite images," *IEEE Trans. Image Process.*, vol. 20, no. 7, pp. 2017–2029, Jul. 2011.

[12] A. Borji and L. Itti, "State-of-the-art in visual attention modeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 185–207, Jan. 2013.

[13] N. D. B. Bruce and J. K. Tsotsos, "Saliency based on information maximization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2005, pp. 155–162.

[14] L. Itti and P. Baldi, "Bayesian surprise attracts human attention," *Vis. Res.*, vol. 49, no. 10, pp. 1295–1306, Jun. 2009.

[15] X. Hou and L. Zhang, "Saliency detection: A spectral residual approach," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2007, pp. 1–8.

[16] Y. Yu, B. Wang, and L. Zhang, "Pulse discrete cosine transform for saliency-based visual attention," in *Proc. 8th IEEE Int. Conf. Develop. Learn.*, Jun. 2009, pp. 1–6.

[17] M.-M. Cheng, N. J. Mitra, X. Huang, P. H. S. Torr, and S.-M. Hu, "Global contrast based salient region detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 569–582, Mar. 2015.

[18] Y. Yu, Z. H. Ding, B. Wang, and L. M. Zhang, "Visual attention-based ship detection in SAR images," in *Advances in Neural Network Research and Applications*, vol. 67. Berlin, Germany: Springer, 2010, pp. 283–292.

[19] S. Liu, Z. J. Cao, and J. Li, "A SVD-based visual attention detection algorithm of SAR image," in *Proc. 2nd Int. Conf. Commun., Signal Process., Syst.*, 2014, pp. 479–486.

[20] B. Hou, W. Yang, S. Wang, and X. Hou, "SAR image ship detection based on visual attention model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 2003–2006.

[21] G. Gao, "An improved scheme for target discrimination in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 277–294, Jan. 2011.

[22] R. Adams and L. Bischof, "Seeded region growing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 16, no. 6, pp. 641–647, Jun. 1994.

[23] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 27:1–27:27, 2011.

[24] M. C. Burl, G. J. Owirka, and L. M. Novak, "Texture discrimination in synthetic aperture radar imagery," in *Proc. IEEE 23rd Asilomar Conf. Signals, Syst., Comput.*, Oct./Nov. 1989, pp. 399–404.

[25] S. M. Verbout, A. L. Weaver, and L. M. Novak, "New image features for discriminating targets from clutter," *Proc. SPIE*, vol. 3395, pp. 120–137, May 1998.

[26] Y. Lin and B. Bhanu, "Evolutionary feature synthesis for object recognition," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 35, no. 2, pp. 156–171, May 2005.

[27] B. Bhanu and Y. Q. Lin, "Genetic algorithm based feature selection for target detection in SAR images," *Image Vis. Comput.*, vol. 21, no. 7, pp. 591–608, 2003.

[28] G. Gao, "The research on automatic acquirement of target's ROI from SAR imagery," Ph.D. dissertation, Nat. Univ. Defense Technol., Changsha, China, 2007.

[29] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[30] J. Singh and M. Datcu, "SAR image categorization with log cumulants of the fractional Fourier transform coefficients," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 12, pp. 5273–5282, Dec. 2013.

[31] G. Dong, N. Wang, and G. Kuang, "Sparse representation of monogenic signal: With application to target recognition in SAR images," *IEEE Signal Process. Lett.*, vol. 21, no. 8, pp. 952–956, Aug. 2014.

[32] S. Chen and H. Wang, "SAR target recognition based on deep learning," in *Proc. IEEE Int. Conf. Data Sci. Adv. Anal.*, Oct./Nov. 2015, pp. 541–547.

[33] S. Tu and Y. Su, "Fast and accurate target detection based on multiscale saliency and active contour model for high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5729–5744, Oct. 2016.

[34] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn. (ICML)*, 2008, pp. 1096–1103.

[35] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, 2006.

[36] P. Tait, *Introduction to Radar Target Recognition* (IEE Radar Series). Stevenage, U.K.: Institution Engineering Technology, 2005.

[37] C. Clemente, L. Pallotta, D. Gaglione, A. De Maio, and J. J. Soraghan, "Automatic target recognition of military vehicles with Krawtchouk moments," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 1, pp. 493–500, Feb. 2017.

[38] C. Clemente, L. Pallotta, I. Proudler, A. De Maio, J. J. Soraghan, and A. Farina, "Pseudo-Zernike-based multi-pass automatic target recognition from multi-channel synthetic aperture radar," *IET Radar, Sonar Navigat.*, vol. 9, no. 4, pp. 457–466, Apr. 2015.

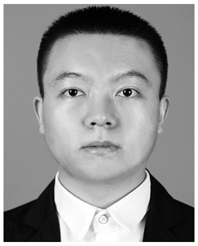
[39] Q. Zhao and J. C. Principe, "Support vector machines for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 643–654, Apr. 2001.

[40] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR detection algorithm for generalized gamma distributed background in high-resolution SAR images," *IEEE Geosci. Remote Sens. Let.*, vol. 10, no. 4, pp. 806–810, Jul. 2013.

[41] K. El-Darymli, P. McGuire, D. Power, and C. R. Moloney, "Target detection in synthetic aperture radar imagery: A state-of-the-art survey," *J. Appl. Remote Sens.*, vol. 7, no. 1, pp. 071598-1–071598-35, Apr. 2013.

[42] F. Dellinger, J. Delon, Y. Gousseau, J. Michel, and F. Tupin, "SAR-SIFT: A SIFT-like algorithm for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 453–466, Jan. 2015.

[43] J. C. Curlander, "Utilization of spaceborne SAR data for mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-22, no. 2, pp. 106–112, Mar. 1984.



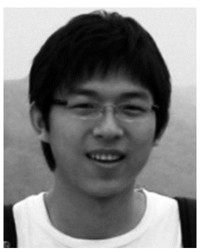
Zhaocheng Wang received the B.S. degree in electronic information science and technology from the Qingdao University of Technology, Qingdao, China, in 2013. He is currently pursuing the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

His research interests include synthetic aperture radar (SAR) automatic target recognition, SAR image processing, and pattern recognition.

Lan Du (M'11–SM'17) received the B.S., M.S., and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 2001, 2004, and 2007, respectively.

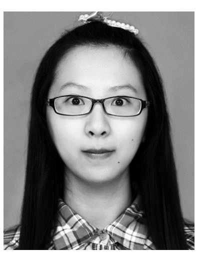
From 2007 to 2009, she was a Post-Doctoral Research Associate with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. She is currently a Professor with the National Laboratory of Radar Signal Processing, Xidian University. Her research interests include statistical signal processing and machine learning with application to radar target recognition.

Dr. Du's doctoral dissertation was granted the National Excellent Doctoral Dissertation of China in 2009. Her research work is supported by the National Natural Science Foundation of China for Excellent Young Scholars and Chang Jiang Scholars Program for Young Scholars.



Peng Zhang (M'14) received the B.S. degree in electronic and information engineering and the M.S. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, China, in 2006, 2009, and 2012, respectively.

He is currently an Associate Professor with the National Laboratory of Radar Signal Processing, Xidian University. His research interests include synthetic aperture radar image analysis and interpretation and statistical learning theory.



Lu Li received the B.S. degree in electronic and information engineering from Xidian University, Xi'an, China, in 2015, where she is currently pursuing the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing.

Her research interests include synthetic aperture radar automatic target recognition and pattern recognition.

Fei Wang received the B.S. and M.S. degrees in electronic engineering from Xidian University, Xi'an, China, in 2008 and 2015, respectively.

Her research interests include radar automatic target recognition, image processing, and linear classifier design.

Shuwen Xu (M'15) received the B.S. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 2006 and 2011, respectively.

He is currently an Associate Professor with the National Laboratory of Radar Signal Processing, Xidian University. His research interests include radar target detection, time–frequency analysis, and synthetic aperture radar image processing.

Hongtao Su received the B.S., M.S., and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 1997, 2000, and 2005, respectively.

He is currently a Professor with the National Laboratory of Radar Signal Processing, Xidian University. His research interests include high-frequency over-the-horizon radar signal processing, adaptive array signal processing, and statistical signal processing.